

Biomimetic Anchors for Antifouling and Antibacterial Polymer Brushes on Stainless Steel

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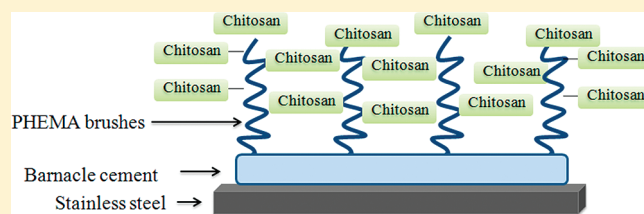
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ABSTRACT: Barnacle cement (BC) was beneficially applied on stainless steel (SS) to serve as the initiator anchor for surface-initiated polymerization. The amine and hydroxyl moieties of barnacle cement reacted with 2-bromoisobutryl bromide to provide the alkyl halide initiator for the surface-initiated atom transfer radical polymerization (ATRP) of 2-hydroxyethyl methacrylate (HEMA). The hydroxyl groups of HEMA polymer (PHEMA) were then converted to carboxyl groups for coupling of chitosan (CS) to impart the SS surface with both antifouling and antibacterial properties. The surface-functionalized SS reduced bovine serum albumin adsorption, bacterial adhesion, and exhibited antibacterial efficacy against *Escherichia coli* (*E. coli*). The effectiveness of barnacle cement as an initiator anchor was compared to that of dopamine, a marine mussel inspired biomimetic anchor previously used in surface-initiated polymerization. The results indicate that the barnacle cement is a stable and effective anchor for functional surface coatings and polymer brushes.



1. INTRODUCTION

The adherence of microorganisms to material surfaces is of crucial importance in biomedical and biomaterial applications. Once bacteria adhered to a surface, the biofilm develops quickly and exhibits extreme resistance to antibiotics and host defense system, leading to devastation and failure of the biomedical devices and biomaterials.^{1,2} In order to prevent and reduce bacterial adhesion, as well as biofilm formation, altering the surface properties via surface modification is essential to maintaining the efficient performance and increasing the service life of biomaterials.^{3,4} Tethering of functional polymer brushes is an effective means to impart antifouling and antibacterial properties to the biomaterials.^{5–8} Two strategies are well-known. One is to immobilize polymers from solution to the surface (the “grafting-to approach”) by physisorption and in many instances covalent bonding. The other approach is surface-initiated (SI) polymerization of monomers from substrates with surface-bound initiators (the “grafting-from approach”). The latter approach allows the preparation of dense and long polymer brushes.^{9–12} For the “grafting-from” method, the presence of a stable and durable layer of initiators on the surface is critical for the production of well-defined polymer brushes to be used in long-term applications.^{7,9,10}

Polymerization initiators have been immobilized on substrates via the use of chemical coupling agents, such as bromomethyl- or chloromethyl-terminated silanes^{13–16} and bromomethyl-terminated thiol agents.^{17–20} To enhance the environmental friendliness, biological adhesives are a good alternative to the traditional chemical coupling agents. They are nontoxic and stable, and they exhibit strong adhesion to a variety of substrates.^{21,22} Among the diverse organisms that produce and utilize biological adhesives, mussels and barnacles have elicited considerable scientific attention. Dopamine, inspired by mussel adhesive, has been used successfully as a biomimetic anchor for SI polymerization in surface modification chemistry.^{23–26} On the other hand, other biomimetic anchors, such as those inspired by barnacle adhesive or the original barnacle cement, have yet to be explored for surface modification. Barnacles attach to a wide variety of substrates, such as metals, minerals, synthetic polymers, biotic surfaces, and even their own calcareous base.²¹ Apart from its outstanding adhesive strength^{29–32} as well as good tensile and shear resistance,^{33,34} the barnacle cement exhibits excellent sustainability as barnacles rarely move once they

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have attached to a substrate.^{21,28} Furthermore, barnacle cement is a proteinaceous adhesive, consisting of commonly known amino acids.²⁸ Last but not least, barnacle cement is an “intelligent” adhesive that uses different adhesive motifs on different substrates.^{21,35,36} The “intelligent” property is very important for efficient bonding to different substrates. Thus, barnacle cement shows considerable potential to be used as an initiator anchor for surface-initiated (SI) polymerization.

In this study, barnacle cement was beneficially applied as an initiator anchor for SI polymerization, enabling growth of polymer brushes with antifouling and antibacterial properties on stainless steel (SS). Barnacle cement anchored on the SS surface was functionalized with alkyl bromide initiator, followed by SI atom transfer radical polymerization (ATRP) of 2-hydroxyethyl methacrylate (HEMA) to introduce the antifouling HEMA polymer (PHEMA) brushes.^{7,37,38} Chitosan (CS), a widely used biocompatible natural biopolymer with antibacterial properties,^{39–41} was subsequently coupled to the PHEMA brushes to impart antibacterial property to the surface. For comparison purpose, dopamine was also employed as the anchor for polymerization initiators in the same surface functionalization process. The antifouling and antibacterial efficacy of the modified SS substrates were assayed with bovine serum albumin (BSA) adsorption and *Escherichia coli* (*E. coli*) adhesion, respectively.

2. MATERIALS AND METHODS

2.1. Materials. AISI type 304 stainless steel foils of 0.05 mm in thickness were purchased from Goodfellow Ltd. of Cambridge, UK. 2-Hydroxyethyl methacrylate (HEMA, 97%) and chitosan (CS, low molecular weight, 75–85% deacetylated) were obtained from Sigma-Aldrich Co., St. Louis, MO. HEMA was passed through an inhibitor-removal column (Sigma-Aldrich) and then stored in argon at -10°C . Dopamine hydrochloride (98%), triethylamine (TEA, 98%), 2-bromoisobutyl bromide (BIBB, 98%), copper(I) bromide (CuBr, 99%), copper(II) bromide (CuBr₂, 99%), *N,N,N',N',N''*-pentamethyldiethylenetriamine (PMDETA, reagent grade), succinic anhydride (SA, 97%), 4-(dimethylamino)pyridine (DMAP, 99%), *N*-hydroxysuccinimide (NHS, 98%), and 1-ethyl-3-(3-(dimethylamino)propyl)carbodiimide hydrochloride (EDAC, purum, $\geq 98.0\%$) were obtained from Sigma-Aldrich Co., St. Louis, MO. CuBr was purified by stirring in acetic acid for 4 h, followed by washing with ethanol and diethyl ether prior to being stored in argon. Solvents, such as dichloromethane, dioxane, ethanol, and acetone, were of analytical grade and were used as received. Bovine serum albumin (BSA, 98%) was also obtained from Sigma-Aldrich Co., St. Louis, MO. *Escherichia coli* (*E. coli*, ATCC, 14948) was obtained from American Type Culture Collection, Manassas, VA. Nutrient broth (containing 1 g/L of beef extract, 5 g/L of peptone, 5 g/L of sodium chloride, and 2 g/L of yeast extract, and with a final pH of 7.4 ± 0.2) was purchased from Sigma-Aldrich Co., St. Louis, MO. The LIVE/DEAD BacLight Bacterial Viability Kit L131152 was purchased from Molecular Probes Inc., Eugene, OR.

2.2. Immobilization of Biological Adhesives and Atom Transfer Radical Polymerization (ATRP) Initiators on Stainless Steel (SS). The SS foils were cut into 1 cm $\times$ 1 cm coupons and cleaned ultrasonically with deionized water, acetone, and ethanol for 5 min each. The SS coupons were then rinsed thoroughly with deionized water and blown dry with argon. The clean SS foils were activated by immersing in the piranha solution (H_2SO_4 (95–97%): H_2O_2 (30%) = 3:1, v:v) for 30 min to generate a hydroxyl-enriched surface, denoted as the SS-OH surface.⁴²

Uncured barnacle cement (BC) was harvested from live barnacles *Amphibalanus amphitrite*⁴² collected from Kranji mangroves, Singapore,

according to the method described by Dickinson et al.²⁷ Briefly, the shell plates (including the base plate) of barnacles were gently cleaned with deionized water and a cotton swab. The barnacles were then dried in air for 3 h. The periphery of the base plate was gently picked in an outward direction with a dissecting needle. Droplets (1–2 μL) of barnacle cement secreted from the adhesive channels were then collected using a pipettor fitted with a microtip. The barnacle cement was deposited directly on the SS foil and then smeared evenly over the surface. The barnacle cement was allowed to cure in air on the stainless steel surface for 15 min. The surface was then washed three times with deionized water and dried under reduced pressure. The barnacle cement-coated surface is referred to as the SS-BC surface.

For the immobilization of dopamine on stainless steel, dopamine was dissolved in 10 mM Tris-HCl (pH 8.5) to a concentration of 2 mg/mL, and the SS-OH substrates were immersed in the solution for 24 h.²⁴ The reaction mixture was stirred thoroughly to prevent nonspecific micro-particle deposition on the surfaces. After the reaction, the substrates were removed, rinsed with deionized water, and dried under reduced pressure. Dopamine undergoes self-polymerization to produce a polydopamine (PDA) layer that anchors strongly to the substrate. The polydopamine-coated surface was denoted as the SS-PDA surface.

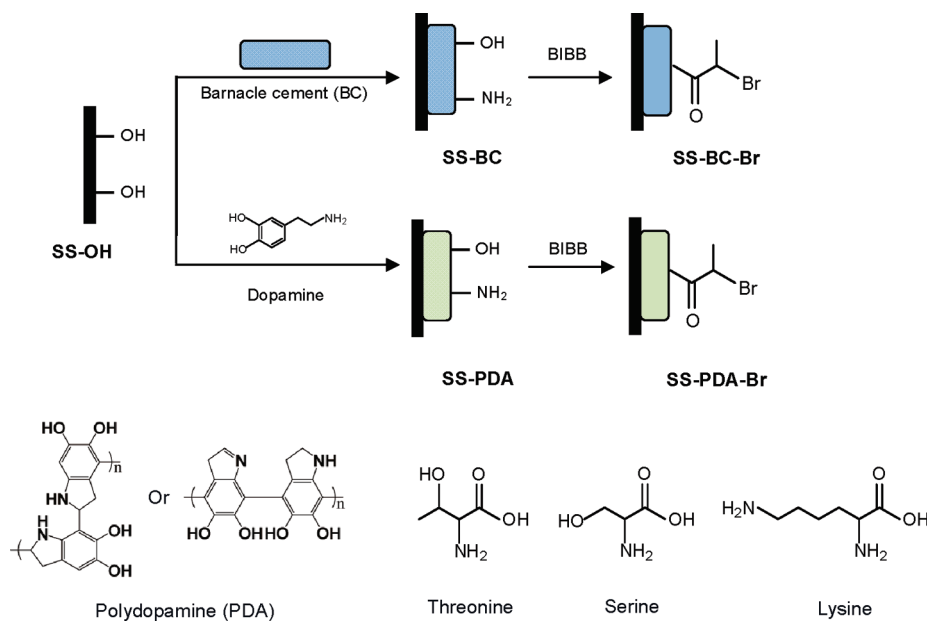
To introduce alkyl bromide onto the substrates as the ATRP initiator, the SS-BC (or SS-PDA) substrate was immersed in 20 mL of dichloromethane containing 2.0 mL (14.4 mmol) of TEA. 2-Bromoisobutyl bromide (1.8 mL, 3.34 g, 14.4 mmol) in 10 mL of dichloromethane was added dropwise with the reaction mixture cooled in an ice bath. The reaction was allowed to proceed at room temperature with stirring for 24 h. The resulting surfaces, referred to as the SS-BC-Br surface or SS-PDA-Br surface, were each rinsed three times with 10 mL of acetone, ethanol, and deionized water, in that order, and then stored in a vacuum desiccator after being dried under reduced pressure overnight.

2.3. Surface-Initiated ATRP of 2-Hydroxyethyl Methacrylate (HEMA) and Immobilization of Chitosan. For the preparation of HEMA polymer (PHEMA) brushes on the SS-BC-Br (or SS-PDA-Br) surfaces, surface-initiated ATRP of HEMA was carried out at a [HEMA]:[CuBr]:[CuBr₂]:[PMDETA] molar feed ratio of 100:1:0.2:1 in 4 mL of deionized water at room temperature in a 10 mL round-bottom flask containing the SS-BC-Br (or SS-PDA-Br) coupons. The mixture was stirred and degassed with argon for 20 min prior to the addition of PMDETA. The reaction was allowed to proceed for 2 h to give rise to the HEMA polymer-grafted surfaces (SS-BC-PHEMA or SS-PDA-PHEMA). After the reaction, the SS-BC-PHEMA (or SS-PDA-PHEMA) surface was washed thoroughly with ethanol and deionized water. The substrate was subsequently immersed in 20 mL of ethanol for about 48 h to ensure the complete removal of the adhered and physically adsorbed polymer. The solvent was changed every 8 h.

To immobilize chitosan on the surfaces, the hydroxyl groups of PHEMA on the SS-BC-PHEMA (or SS-PDA-PHEMA) substrate were first converted into carboxyl ($-\text{COOH}$) groups for the reaction with the amine ($-\text{NH}_2$) groups in chitosan.⁴⁴ Succinic anhydride (SA, 0.1 g) was dissolved in 8 mL of dioxane, and the SS-BC-PHEMA (or SS-PDA-PHEMA) coupon was introduced into the solution. Triethylamine (TEA, 0.3 mL) and 4-(dimethylamino)pyridine (DMAP, 0.12 g) were added to initiate the reaction. The reaction was allowed to proceed for 24 h at room temperature to produce the SS-BC-SA (or SS-PDA-SA) surface. After the reaction, the substrate was washed thrice with 20 mL of ethanol and deionized water to remove the adsorbed reagents.

To increase the reactivity with chitosan (CS), the $-\text{COOH}$ groups on the SS-BC-SA (or SS-PDA-SA) surface was preactivated for 1 h at room temperature in phosphate buffered saline (PBS) solution, containing 1 mg/mL of NHS and 10 mg/mL of EDAC. The substrates were then washed with deionized water and transferred to the chitosan solution (1.5 mg/mL in 0.1 M acetic acid solution). The reaction was allowed to proceed for 24 h to produce the SS-BC-CS (or SS-PDA-CS)

Scheme 1. Immobilization of ATRP Initiator via Barnacle Cement and Dopamine



surface. The resulting substrates were each rinsed with 50 mL of deionized water and dried under reduced pressure.

2.4. Surface Characterization. The composition of the surface-functionalized substrates was determined by X-ray photoelectron spectroscopy (XPS). The XPS measurements were performed on a Kratos AXIS Ultra HSA spectrometer with a monochromatized Al K α X-ray source (1486.6 eV photons), at a constant dwelling time of 100 ms and pass energy of 40 eV. The core-level signals were obtained at the photoelectron takeoff angle (α , with respect to the sample surface) of 90°. All binding energies (BEs) were referenced to the C 1s hydrocarbon peak at 284.6 eV. In peak analysis, the line width (full width at half-maximum, or fwhm) for the Gaussian peaks was maintained constant for all peak components in a particular spectrum. Surface elemental stoichiometries were determined from peak-area ratios, after correcting with the experimentally determined sensitivity factors, and were reliable to $\pm 5\%$.

Static water contact angles of the pristine and surface functionalized SS substrates were measured at 25 °C, using the sessile drop method with a 2 μ L water droplet, in a telescopic goniometer (model 100-00-(230), Rame-Hart, Inc., Mountain Lake, NJ). Each static water contact angle was the mean value from three substrates, with the value of each substrate obtained by averaging the contact angles from at least three surface locations.

2.5. Protein Adsorption Test. Protein adsorption of the modified surfaces was assayed by immersion of the substrates in phosphate buffered saline (PBS) of BSA⁴⁵ and sterilized nutrient broth for cell culture. Pristine SS, SS-BC, SS-PDA, SS-BC-PHEMA, SS-PDA-PHEMA, SS-BC-CS, and SS-PDA-CS substrates were rinsed initially with PBS to rehydrate the surfaces, prior to being placed in the 2 mg/mL PBS of BSA and the sterilized nutrient broth. The adsorption was allowed to proceed at 37 °C for 24 h. Substrates were then removed from the solution, gently washed four times with PBS, and rinsed twice with deionized water to remove PBS salts. After drying under reduced pressure, the protein-sorped surfaces were characterized by XPS. The intensity ratio of the N 1s signal from the peptide bonds to the total C 1s signal is employed as an indicator for comparing the relative amounts of the protein adsorbed on the substrates.^{45,46}

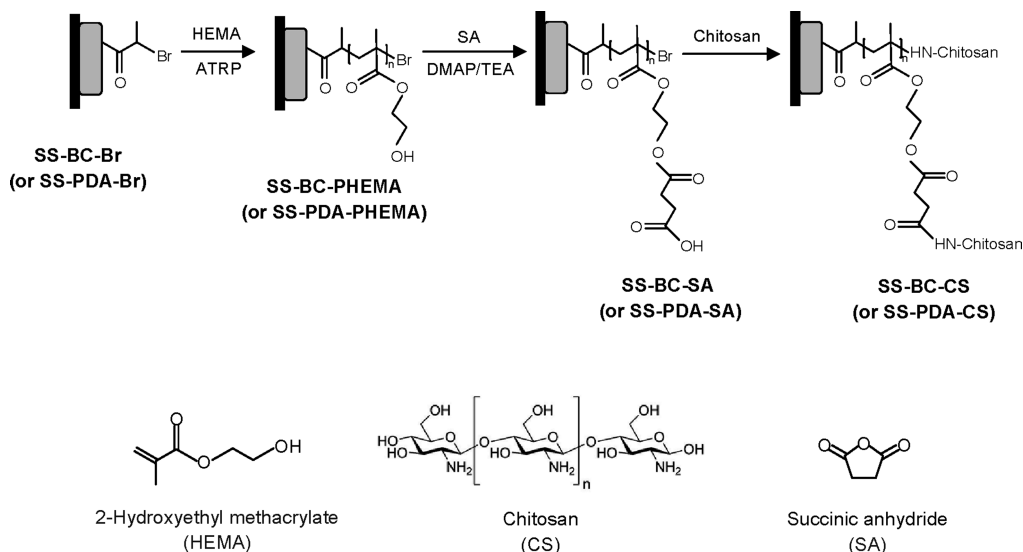
2.6. Bacterial Assays. The live/dead two-color fluorescence method was used to assess bactericidal effect of functionalized substrates on Gram-negative *E. coli*.^{38,47} The LIVE/DEAD BacLight Bacterial

Viability Kit consists of a mixture of SYTO 9 green fluorescent nucleic acid dye and propidium iodide (PI) red fluorescent nucleic acid dye. SYTO 9 is membrane permeable and therefore stains both viable and nonviable bacteria. PI, which has a higher affinity for nucleic acids, is rejected from viable bacterial cells by membrane pumps. When both dyes are present, PI competes with SYTO 9 for nucleic acid binding sites. Thus, viable (appearing green) and dead (appearing red) bacterial cells can be distinguished with fluorescent microscopy.

The bacterial used for the antibacterial assays was cultivated in the nutrient broth at 37 °C for 18 h. The bacteria-containing broth was centrifuged at 2700 rpm for 10 min. After the removal of the supernatant, the bacterial cells were washed twice with PBS and then resuspended in PBS at a concentration of 10^6 cells/mL. Bacterial cell concentration was estimated by measuring the absorbance of cell dispersions at 540 nm and referenced to a standard calibration curve. An optical density of 1.0 at 540 nm is approximately equivalent to 10^9 cells/mL.⁴⁷ All glassware was sterilized in an autoclave at 120 °C for 20 min. The pristine and functionalized stainless steel substrates were sterilized with UV irradiation for 1 h before the experiment. Each substrate was immersed in 5 mL of the above bacterial suspension in PBS at a concentration of 10^6 cells/mL at 37 °C. After 4 h, the substrates were gently washed twice with PBS to remove the loosely attached cells and subsequently stained by dropping 0.1 mL solution of the LIVE/DEAD BacLight Kit on the substrate surfaces for 15 min. The stained substrates were viewed under a green filter (excitation/emission, 420–480 nm/520–580 nm) and a red filter (excitation/emission, 480–550 nm/590–800 nm) with a Leica DMLM microscope equipped with a 100 W Hg lamp. At least three different surface locations on each substrate were randomly chosen for imaging to obtain a representative overview of the surface.

Quantification of bacterial adhesion and viability on the pristine and functionalized SS was carried out by the spread plate method.^{39,40} The samples were immersed in 1 mL of bacterial suspension in PBS with an initial cell concentration of 10^8 cells/mL in 24-well plates at 37 °C. After 4 h, the substrates were gently washed thrice with sterile PBS and then cleaned in 2 mL of sterile PBS solution under mild ultrasonic for 5–7 min. The bacterial suspension was mixed in a vortex mixer for 15 s, followed by 10-fold serial dilution. 100 μ L aliquots of the serially diluted suspension were spread onto the triplicate solid agar. After incubation of the plates at 37 °C for 24 h, the number of viable cells (colonies) was

Scheme 2. Surface-Initiated Atom Transfer Radical Polymerization of HEMA and Immobilization of Chitosan



counted manually, and the results were expressed as the relative viability, defined as the percentage of viable cells on the modified SS substrates relative to that on the pristine SS.

2.7. Stability of the Biomimetic Anchors. To assess the stability of barnacle cement and dopamine as initiator anchors on SS surfaces, the SS-BC and SS-PDA substrates were immersed in sterile PBS at 37 °C for 30 days. After the immersion, the substrates were washed with PBS and deionized water before being dried under reduced pressure. The composition of each substrates surface was then determined by XPS analysis.

3. RESULTS AND DISCUSSION

3.1. Surface Modification of Stainless Steel Substrates. The procedures for the functionalization of stainless steel (SS) surfaces with chitosan-coupled HEMA polymer (PHEMA) brushes using barnacle cement or dopamine as the biomimetic anchor are shown in Schemes 1 and 2. Barnacle cement or dopamine was first coated on SS for the subsequent reaction with 2-bromoisobutyryl bromide to introduce the alkyl bromide initiator for surface-initiated (SI) atom transfer radical polymerization (ATRP). PHEMA brushes were grafted from the surface via SI ATRP of 2-hydroxyethyl methacrylate (HEMA) and subsequently reacted with succinic anhydride to produce carboxyl groups on the surface. Chitosan was finally coupled to the PHEMA brushes through reaction of the amine groups in the former with the carboxyl groups in the latter to give rise to the bifunctional (antifouling and antibacterial) surfaces.

3.2. Bifunctional SS Surfaces Prepared Using Barnacle Cement as a Biomimetic Anchor. Barnacle cement is composed of ~90% protein with residual levels of carbohydrate (1%), lipid (1%), and inorganic ash (4%).⁴⁹ The amino acids threonine, serine, and lysine are known to be abundant in barnacle cement.^{28,50} The hydroxyl groups in serine and threonine as well as amine groups in lysine are very reactive and can react with 2-bromoisobutyryl bromide to introduce the alkyl halide ATRP initiator (Scheme 1). The subsequent surface-initiated atom transfer radical polymerization would allow the preparation of well-defined polymer brushes.^{7,13,14}

The respective X-ray photoelectron spectroscopy (XPS) wide scan and C 1s core-level spectra of the barnacle cement-coated SS (SS-BC surface, Figure 1a,b) and the surfaces after each step of functionalization (Figure 1c–j) are shown in Figure 1. The XPS signals of the metallic elements from underlying SS (Fe 2p_{3/2} and Cr 2p_{3/2} core-level signals with BEs at about 711 and 577 eV, respectively) are not discernible in the wide scan spectra of all functionalized surfaces, indicating the complete coverage of underlying metal substrates. The corresponding static water contact angle and composition of each surface are listed in Table 1. After pretreatment with piranha solution, the static water contact angle of SS decreases from 78 ± 2° to 41 ± 3° (Table 1), indicating a more hydrophilic surface with the introduction of hydroxyl (–OH) groups.

Barnacle cement was extracted from the live adult barnacles and used as its native state on the SS surfaces. The barnacle cement-modified SS surface is hydrophobic with a static water contact angle of 89 ± 4° (Table 1). A strong N 1s signal, associated with the peptide bonds and amino acids, is observed in the XPS wide scan spectrum of the SS-BC surface (Figure 1a). The C 1s core-level spectrum of the SS-BC surface (Figure 1b) can be curve-fitted into five peak components with binding energies (BEs) at about 284.6, 285.6, 286.2, 287.8, and 288.9 eV, attributable to the C–H, C–N, C–O, N–C=O, and O–C=O species, respectively.^{46,51} The [N]/[C] ratio, derived from the XPS N 1s and C 1s core-level spectral area ratio, is 0.24 for the SS-BC surface (Table 1). After the barnacle cement has reacted with 2-bromoisobutyryl bromide, the static water contact angle decreases to 69 ± 3° (Table 1). Concomitantly, three additional peak components with BEs at about 70, 189, and 256 eV, associated with the Br 3d, Br 3p, and Br 3s core-level signals, respectively,⁵¹ appear in the wide scan spectrum of the SS-BC-Br surface (Figure 1c). This result indicates successful introduction of the alkyl bromide-containing ATRP initiator onto the SS-BC surface. The C 1s core-level spectrum of the SS-BC-Br surface (Figure 1d) can be curve-fitted into five peak components with BEs at 284.6, 285.6, 286.2, 287.8, and 288.9 eV, attributable to C–H, C–N, C–O/C–Br, N–C=O, and O–C=O species, respectively. The corresponding Br 3d core-level spectrum of the

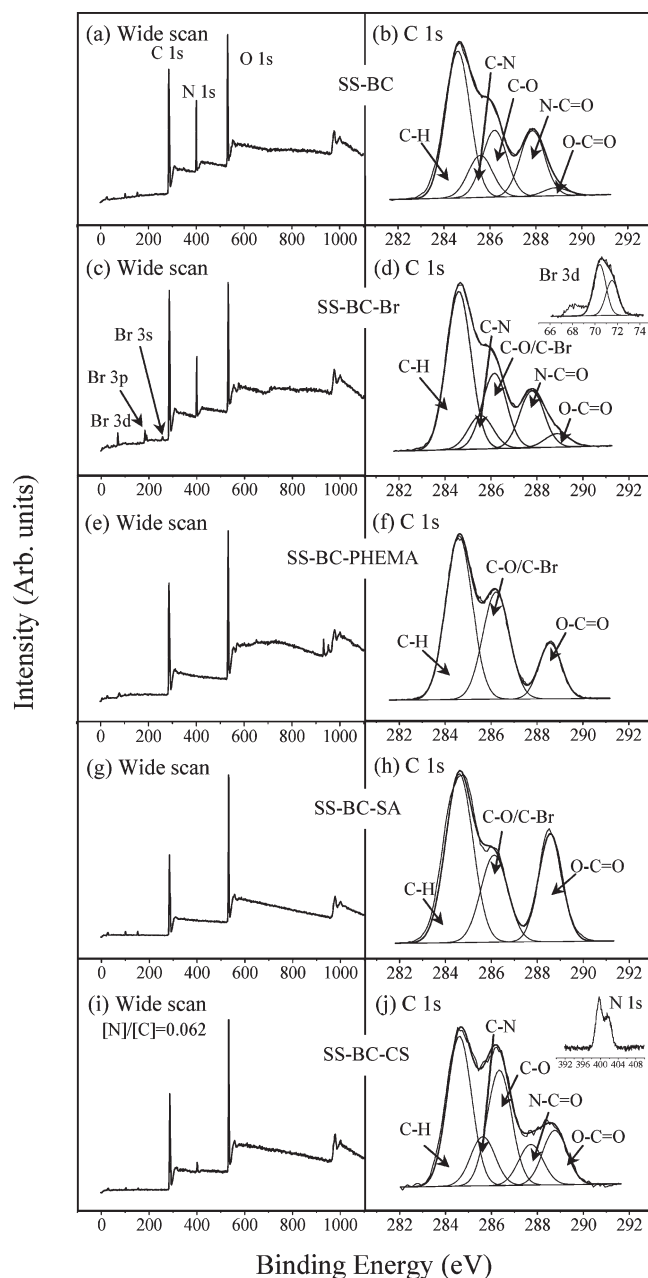


Figure 1. XPS wide scan and C 1s core-level spectra of (a, b) the SS-BC surface, (c, d) the SS-BC-Br surface (inset: Br 3d core-level spectrum), (e, f) the SS-BC-PHEMA, (g, h) the SS-BC-SA, and (i, j) the SS-BC-CS surfaces (inset: N 1s core-level spectrum).

SS-BC-Br surface, shown as an inset in Figure 1d, is curve-fitted into two peak components with an area ratio of about 3:2. The Br $3d_{5/2}$ and Br $3d_{3/2}$ peak components with respective BEs at 70.4 and 71.5 eV are associated with the covalently bonded alkyl bromide species on the surface.⁵¹

After the surface-initiated polymerization of 2-hydroxyethyl methacrylate, the static water contact angle of the SS-BC-PHEMA surface decreases to $45 \pm 2^\circ$ (Table 1), in agreement with the hydrophilic nature of PHEMA brushes. In comparison to the wide scan spectrum of the SS-BC-Br surface (Figure 1c), the N 1s signal (at the BE of about 400 eV⁵²) disappears completely in the wide scan spectrum of the SS-BC-PHEMA

surface (Figure 1e), implying that dense PHEMA brushes have been grafted on the SS-BC-Br surface to a thickness exceeding the 8 nm sampling depth of XPS measurements.⁵³ Using experimental conditions similar to ours, the thickness of PHEMA brushes can reach to 26 and 41 nm for 4 and 8 h of SI ATRP.⁵² Thus, the thickness of PHEMA brushes with 2 h of polymerization time used in this work has probably exceeded 10 nm, resulting in the coverage of the N 1s signal from the underlying BC after surface graft polymerization. The C 1s core-level spectrum of the SS-BC-PHEMA surface (Figure 1f) can be curve-fitted into three peak components with BEs at about 284.6, 286.2, and 288.5 eV, attributable to the C-H, C-O/C-Br, and O-C=O species, respectively.^{44,52} The area ratio of the three peak components is about 3.1:2.0:0.9, which is consistent with the theoretical ratio of 3:2:1 for the PHEMA homopolymer.

For the immobilization of chitosan on the SS-BC-PHEMA surface, the pendant hydroxyl groups of PHEMA were first converted into carboxyl groups, via coupling of succinic anhydride (SA), to give rise to the SS-BC-SA surface. The C 1s core-level spectrum of the SS-BC-SA surface (Figure 1h) can be curve-fitted into three peak components with BEs at 284.6, 286.2, and 288.6 eV, which are assigned to the C-H, C-O/C-Br, and O-C=O species, respectively.⁴⁴ In contrast to the C 1s core-level spectrum of SS-BC-PHEMA surface (Figure 1f), the increase in O-C=O peak component (Figure 1h) confirms the presence of carboxyl groups on the SS-BC-SA surface. The peak area ratios of [C-O/C-Br]/[O-C=O] are 2.0:2.2 and 2.0:0.9 for the SS-BC-SA and SS-BC-PHEMA surfaces (Table 1), respectively. From the quantitative changes in peak area ratios, it is determined that about 65% of hydroxyl groups on the PHEMA brushes have been activated. The static water contact angle of the SS-BC-SA surface decreases to $43 \pm 3^\circ$ (Table 1). The carboxyl groups on the SS-BC-SA surface were subsequently reacted with the amine ($-\text{NH}_2$) groups in chitosan (CS) to produce SS-BC-CS surface. The static water contact angle of the SS-BC-CS surface increases slightly to $49 \pm 3^\circ$ (Table 1). In comparison to the wide scan spectrum of the SS-BC-SA surface (Figure 1g), the reappearance of N 1s signal in the wide scan spectrum of SS-BC-CS surface (Figure 1i) suggests the successful immobilization of chitosan. The C 1s core-level spectrum of the SS-BC-CS surface (Figure 1j) can be curve-fitted into five peak components with BEs at about 284.6, 285.6, 286.3, 287.7, and 288.8 eV, attributable to the C-H, C-N, C-O, N-C=O, and O-C=O species, respectively.^{46,52} The obvious increase in intensity of the C-O peak component and decrease in that of the O-C=O peak component are also consistent with chitosan having been coupled to the PHEMA brushes. The N-C=O peak component is associated with the linkage formed between $-\text{COOH}$ groups and $-\text{NH}_2$ groups as well as the peptide linkages in chitosan. The appearance of the N-C=O peak component is thus consistent with the successful immobilization of chitosan on the surface.

3.3. Bifunctional SS Surfaces Prepared Using Dopamine as a Biomimetic Anchor. As a simple molecule that is central to curing of mussel adhesive^{54,55} and mimics the proteins in marine mussel adhesive, dopamine has been recently used as a versatile platform for anchoring functional molecules and polymers onto a variety of surfaces.^{23–26} Dopamine undergoes self-polymerization to produce an adherent polydopamine (PDA) layer on the substrates (Scheme 1). Although the exact polymerization mechanism is not yet clear, the process probably involves the oxidation of catechol to quinone and rearrangement into dihydroxyindole,²⁴ with plenty of hydroxyl and amine groups left on the surfaces. In the

Table 1. Static Water Contact Angles and Surface Composition of the Pristine and Surface-Functionalized Stainless Steel (SS) Substrates

samples		surface composition (molar ratio)	static water contact angle (mean $\pm$ SD, deg)
pristine SS			78 $\pm$ 2
SS-OH			41 $\pm$ 3
SS-BC		[C]:[N] = 4.2:1	89 $\pm$ 4
SS-BC-Br		[C]:[N]:[Br] = 63.7:10.5:1	69 $\pm$ 3
SS-BC-PHEMA		[C-H]:[C-O/C-Br]:[O-C=O] = 3.1:2.0:0.9	45 $\pm$ 2
SS-BC-SA		[C-H]:[C-O/C-Br]:[O-C=O] = 3.9:2.0:2.2	43 $\pm$ 3
SS-BC-CS		[C-H]:[C-N]:[C-O]:[N-C=O]:[O-C=O] = 3.7:1.2:2.8:1.0:1.3	49 $\pm$ 3
SS-PDA		[C]:[N] = 9.1:1.0	63 $\pm$ 2
SS-PDA-Br		[C]:[N]:[Br] = 29.6:1.5:1	66 $\pm$ 3
SS-PDA-PHEMA		[C-H]:[C-O/C-Br]:[O-C=O] = 2.7:2.0:0.9	43 $\pm$ 3
SS-PDA-SA		[C-H]:[C-O/C-Br]:[O-C=O] = 3.5:2.0:2.1	41 $\pm$ 3
SS-PDA-CS		[C-H]:[C-N]:[C-O]:[N-C=O]:[O-C=O] = 3.6:1.0:3.9:1.4:1.2	46 $\pm$ 3

present work, the effectiveness of the barnacle cement as an anchor for antifouling and antibacterial polymer brushes was compared to that of dopamine as a biomimetic anchor for the same polymer brushes. The process for preparation of SS-PDA-CS surface is similar to that for the preparation of SS-BC-CS surface (Schemes 1 and 2).

Figure 2 shows the XPS analysis results of the SS-PDA, SS-PDA-Br, SS-PDA-PHEMA, SS-PDA-SA, and SS-PDA-CS surfaces, while Table 1 lists the corresponding static water contact angles and composition of the surfaces. The total coverage of the substrates by the functional layer results in minimal XPS signals of the underlying metallic elements (Fe 2p_{3/2} and Cr 2p_{3/2} core-level signals with BEs at around 711 and 577 eV, respectively) in the wide scan spectra. The static water contact angle for the SS-PDA surface increases from 41 $\pm$ 3° to 63 $\pm$ 2° (Table 1), consistent with the anchoring of a polydopamine layer on the SS surface. The XPS C 1s core-level spectrum of the SS-PDA surface (Figure 2b) can be curve-fitted into five peak components with BEs at about 284.6, 285.6, 286.2, 287.4, and 288.9 eV, attributable to the C-H, C-N, C-O, C=O, and O-C=O species, respectively.^{46,51} The appearance of the C-N peak component at the BE of 285.6 eV in the C 1s core-level spectrum (Figure 2b) and N 1s peak component at the BE of around 400 eV in the wide scan spectrum of SS-PDA (Figure 2a) confirms the presence of a polydopamine layer on the surface.^{24,56} The [N]/[C] ratio of the SS-PDA surface is 0.11 (Table 1), which is in good agreement of the theoretic value of 0.12 for dopamine. The polydopamine-coated surface provides the hydroxyl and amine groups to allow reaction with 2-bromoisobutyl bromide to introduce the ATRP initiator.^{24,57} As shown in the wide scan spectrum in Figure 2c, the appearance of three additional peaks with BEs at about 70, 189, and 256 eV, associated with Br 3d, Br 3p, and Br 3s core-level signals,⁵¹ respectively, indicates the successful introduction of alkyl bromide-containing ATRP initiator onto the SS-PDA surface. The C 1s core-level spectrum of the SS-PDA-Br surface (Figure 2d) can be curve-fitted into five peak components with BEs at 284.6, 285.6, 286.2, 287.5, and 290.0 eV, attributable to C-H, C-N, C-O/C-Br, N-C=O, and O-C=O species, respectively. The Br 3d core-level spectrum (inset of Figure 2d) can be curve-fitted into two peak components with an area ratio of around 3:2 and with BEs at 70.4 and 71.5 eV, attributable to the respective Br 3d_{5/2} and Br 3d_{3/2} species. The Br 3d spectrum thus confirms the presence of covalently bonded alkyl bromide species on the surface.⁵¹

After the graft polymerization of HEMA on the SS-PDA surface, the modified surface has become hydrophilic with a static water contact angle of 43 $\pm$ 3° (Table 1). In the wide scan spectrum of SS-PDA-PHEMA (Figure 2e), the N 1s signal (at the BE of about 400 eV) has disappeared completely, indicating that a thick layer of PHEMA brushes have been grafted onto the SS-PDA-Br surface. Similar to that of the SS-BC-PHEMA surface, the C 1s core-level spectrum of SS-PDA-PHEMA surfaces (Figure 2f) can be curve-fitted into three peak components with BEs at about 284.6, 286.2, and 288.5 eV, attributable to C-H, C-O/C-Br, and O-C=O species, respectively.^{44,52} The area ratio of the three peaks (2.7:2.0:0.9) is in fairly good agreement with the theoretical ratio of 3:2:1 for the PHEMA homopolymer, consistent with the presence of grafted PHEMA brushes on the surface.

The C 1s core-level spectrum of the SS-PDA-SA surfaces (Figure 2h) can be curve-fitted into three peak components with BEs at 284.6, 286.1, and 288.5 eV, assigned to the C-H, C-O/C-Br, and O-C=O species, respectively.⁴³ In comparison to the C 1s core-level spectrum of the SS-PDA-PHEMA surface (Figure 2f), the intensity of the O-C=O peak component has increased significantly (Figure 2h), indicating the successful conversion of hydroxyl groups into the carboxyl groups. In compared to the area ratio of [C-O/C-Br]/[O-C=O] peak components for the SS-PDA-PHEMA surface (2.0:0.9), the corresponding area ratio for the SS-PDA-SA surface is 2.0:2.1, indicating that about 60% of the hydroxyl groups in the PHEMA brushes have reacted with SA to form the carboxyl groups. The static water contact angle of the SS-PDA-SA surface increases from 41 $\pm$ 3° to 46 $\pm$ 3° (Table 1) after coupling of chitosan. In comparison to the wide scan spectrum the SS-PDA-SA surface (Figure 2g), a strong N 1s signal has reappeared in the wide scan spectrum of SS-PDA-CS surface (Figure 2i), consistent with the presence of immobilized chitosan. The C 1s core-level spectrum for SS-PDA-CS surface (Figure 2j) can be curve-fitted into five peak components with BEs at about 284.6, 285.6, 286.2, 287.7, and 288.7 eV, attributable to the C-H, C-N, C-O, N-C=O, and O-C=O species, respectively.^{46,52} The appearance of N-C=O peak component, the increase in intensity of the C-O peak component, and the decrease in that of the O-C=O peak component confirm that chitosan has been coupled to the PHEMA brushes.

3.4. Protein Adhesion to the Modified SS Surface. XPS has been employed to evaluate the extent of protein adsorption by using the relative intensity of the N 1s core-level signal as a

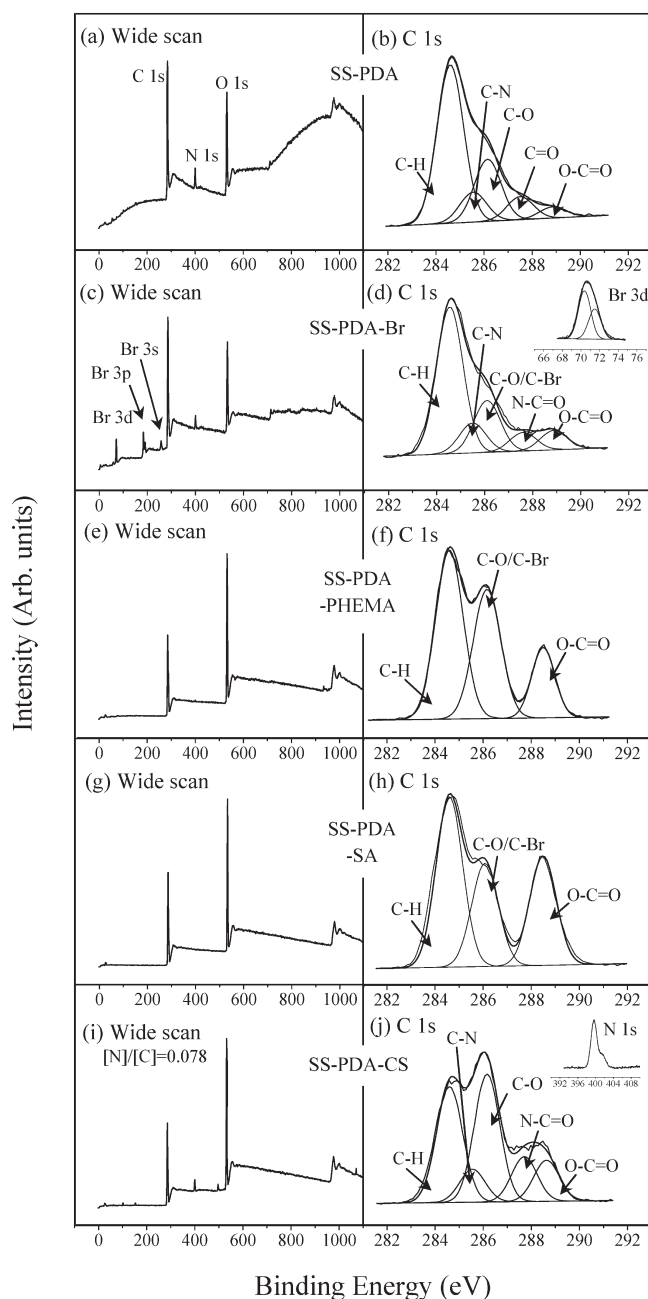


Figure 2. XPS wide scan and C 1s core-level spectra of (a, b) the SS-PDA surface, (c, d) the SS-PDA-Br surface (inset: Br 3d core-level spectrum), (e, f) the SS-PDA-PHEMA, (g, h) the SS-PDA-SA, and (i, j) the SS-PDA-CS surfaces (inset: N 1s core-level spectrum).

marker.^{45,46} The $[N]/[C]$ ratio, determined from the N 1s and C 1s core-level peak area ratio, is used to indicate the relative amount of proteins adsorbed on a surface.^{45,46} The XPS wide scan spectrum of the pristine SS surface before BSA adsorption is shown in Figure 3a, while the respective XPS wide scan and N 1s core-level spectra of the pristine SS, SS-BC, SS-PDA, SS-BC-PHEMA, SS-PDA-PHEMA, SS-BC-CS, and SS-PDA-CS surfaces after 24 h of exposure to the PBS solution of BSA (2 mg/mL) are shown in Figure 3b–h.

In comparison to the wide scan spectrum of the pristine SS surface prior to protein adsorption (Figure 3a), the appearance of a strong N 1s signal at the BE of about 400 eV after BSA exposure

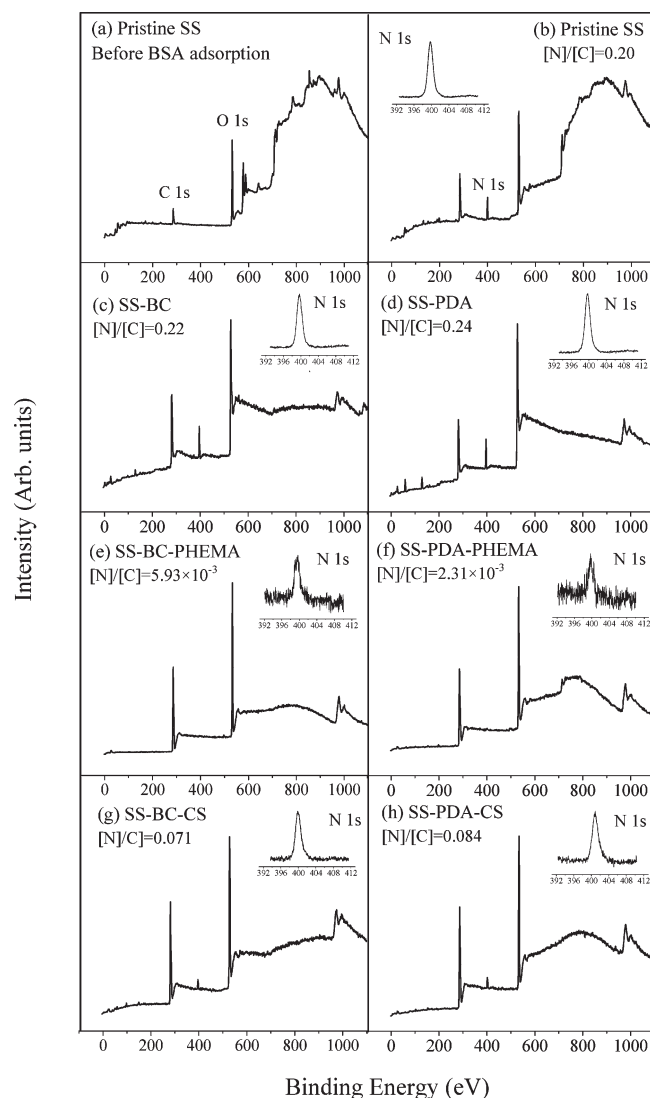


Figure 3. XPS wide scan spectra of (a) the pristine SS before BSA exposure, the (b) pristine SS, (c) SS-BC, (d) SS-PDA, (e) SS-BC-PHEMA, (f) SS-PDA-PHEMA, (g) SS-BC-CS, and (h) SS-PDA-CS surfaces after exposure to 2 mg/mL BSA solution for 24 h.

indicates that extensive protein adsorption has occurred on the pristine SS surface (Figure 3b). The $[N]/[C]$ ratios are 0.0 and 0.20 respectively for the pristine SS surfaces before and after protein adsorption, indicating that the pristine SS surface is susceptible to protein fouling. A broad high BE envelope (700–1000 eV), associated with the underlying metal substrate, appears in the XPS wide scan spectra of pristine SS substrate before and after protein adsorption, indicating that the thickness of adsorbed BSA layer on SS is less than the sampling depth of XPS technique (about 8 nm in an organic matrix^{51,53}). In comparison to the pristine SS surface, the SS-BC and SS-PDA surfaces also exhibit poor antifouling property. The $[N]/[C]$ ratio of SS-BC surface after BSA adsorption is about 0.22 (Figure 3c), which is lower than that of 0.24 for the initial SS-BC surface (see Figure 7b,c). The decrease in $[N]/[C]$ ratio for SS-BC surface after BSA adsorption is consistent with a lower $[N]/[C]$ ratio for the pristine SS surface adsorbed BSA than that of the barnacle cement. The $[N]/[C]$ ratio increases substantially from 0.11 to 0.24 for the SS-PDA surface after exposure to BSA (compare

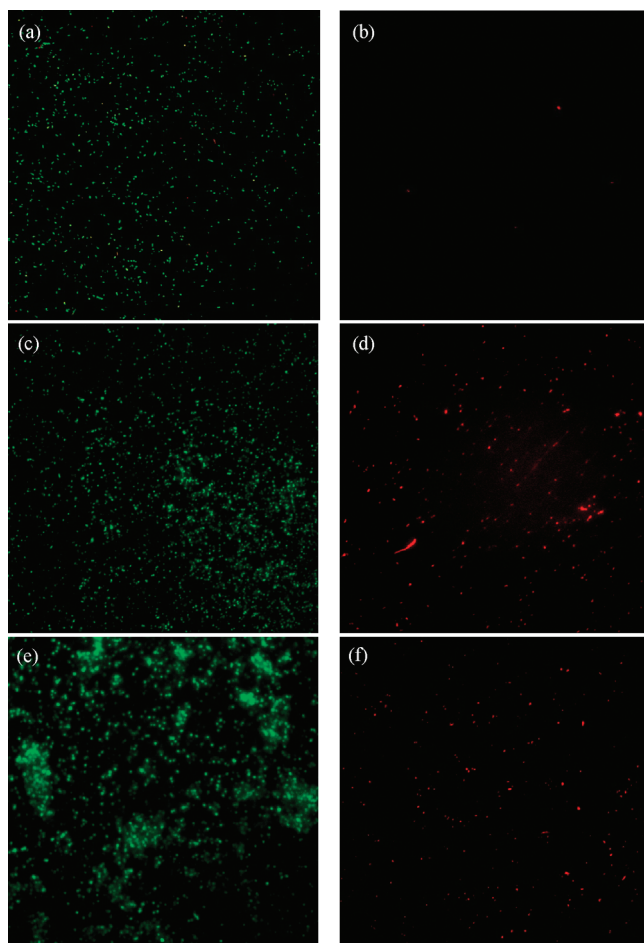


Figure 4. Fluorescence microscopy images of the (a) pristine SS, (c) SS-BC, and (e) SS-PDA surfaces under green filter and the (b) pristine SS, (d) SS-BC, and (f) SS-PDA surfaces under red filter, after immersion in a PBS suspension of *E. coli* (10^6 cells/mL) for 4 h.

Figure 3d to Figure 7h,i), suggesting severe fouling of the SS-PDA surface.

The surfaces modified by poly(HEMA) and chitosan show good antifouling property when exposed to the BSA solution. The N 1s signal (at the BE of about 400 eV) is barely discernible in the wide scan spectra of the SS-BC-PHEMA and SS-PDA-PHEMA surfaces (Figure 3e,f) after exposure to BSA. The corresponding $[N]/[C]$ ratios for SS-BC-PHEMA and SS-PDA-PHEMA surfaces are around 5.93×10^{-3} and 2.31×10^{-3} , respectively. The minimal quantity of adsorbed BSA suggests that the grafted PHEMA brushes prevent protein adsorption. Figure 3g shows the wide scan and N 1s core-level spectra of the SS-BC-CS surface after BSA exposure. No significant changes are observed in the wide scan spectrum of the SS-BC-CS surface after BSA exposure (compare Figure 1i and Figure 3g), and the respective $[N]/[C]$ ratios are about 0.062 and 0.071. The comparable $[N]/[C]$ ratios suggest that the SS-BC-CS surface remains efficient at reducing protein adsorption after chitosan coupling. After exposure to the BSA solution, the $[N]/[C]$ ratio for the SS-PDA-CS surface is 0.084 (Figure 3h), as compared to that of 0.078 prior to BSA exposure (Figure 2i). Thus, the modified SS surfaces using both types of initiator anchors exhibit good protein repellence, demonstrating that both barnacle cement and dopamine are effective initiator anchors for surface-initiated polymerization. This antifouling

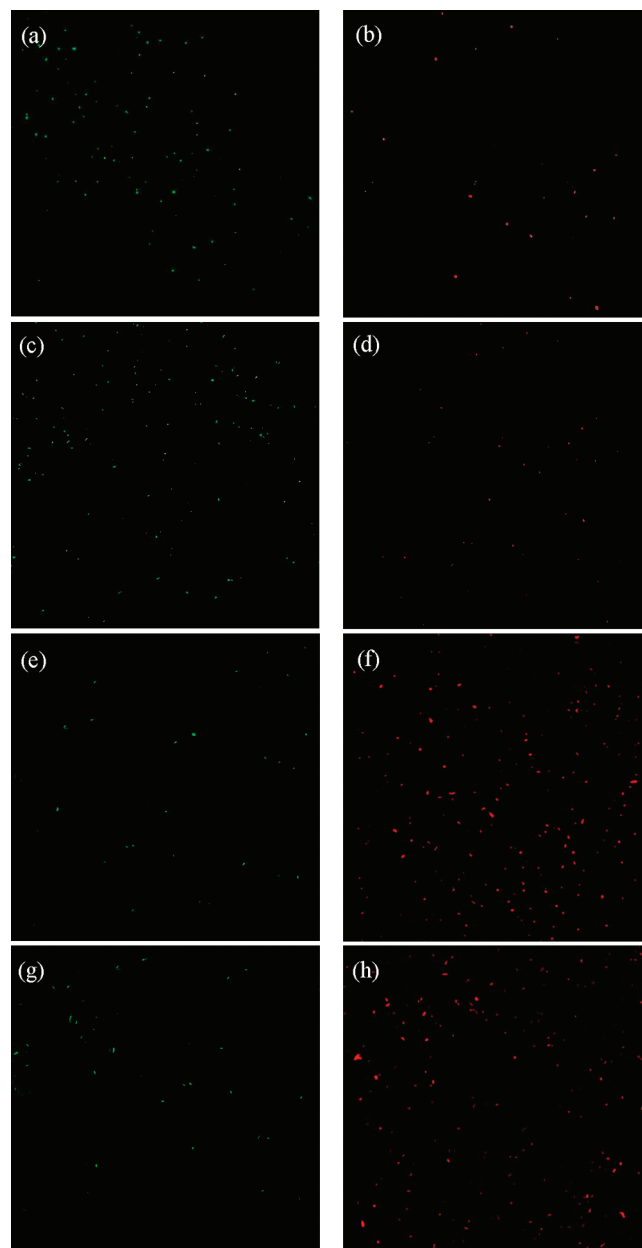


Figure 5. Fluorescence microscopy images of the (a) SS-BC-PHEMA, (c) SS-PDA-PHEMA, (e) SS-BC-CS, and (g) SS-PDA-CS surfaces under green filter and the (b) SS-BC-PHEMA, (d) SS-PDA-PHEMA, (f) SS-BC-CS, and (h) SS-PDA-CS surfaces under red filter, after immersion in a PBS suspension of *E. coli* (10^6 cells/mL) for 4 h.

property is attributed to the surface grafted PHEMA brushes, which become highly extended and oriented to physically exclude protein adsorption and prevent cell adhesion.^{7,59}

3.5. Antibacterial Property of the Modified SS Surface. The antibacterial efficacy of the modified SS surfaces was assayed by both the live/dead two-color fluorescence method and *in vitro* viability test of Gram-negative *E. coli*. Before determining the bacterial adhesion and bactericidal effects, the protein resistance of the modified substrates to the bacterial culture medium (nutrient broth) for *E. coli* was first determined. The XPS-derived $[N]/[C]$ ratios are 0.0 and 0.23 respectively for the pristine SS surfaces before and after 24 h of immersion in the nutrient broth solution,

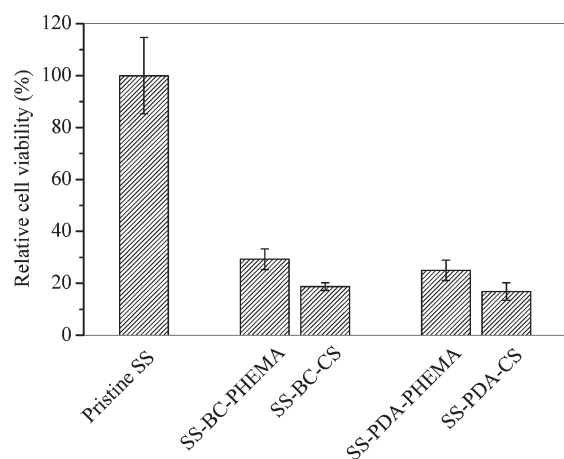


Figure 6. Relative viability of *E. coli* cells in PBS in contact with the pristine and surface-functionalized stainless steel substrates at 37 °C for 4 h. The cell number was determined by the spread plate method. Each error bar represents the standard deviation calculated from three replicates.

while the $[N]/[C]$ ratios for the SS-BC-PHEMA and SS-PDA-PHEMA surfaces are 0.03 and 0.02, respectively, after the immersion. The $[N]/[C]$ ratios for the SS-BC-CS and SS-PDA-CS surfaces after immersion are 0.09 and 0.10, respectively, close to the corresponding values of 0.06 and 0.08 for the SS-BC-CS and SS-PDA-CS substrates prior to immersion. The minimal increase in $[N]/[C]$ ratios for the functionalized SS substrates, compared to that of the pristine SS, suggests good fouling resistance of the surface functionalized substrates in the nutrient broth.

Figures 4 and 5 show fluorescence microscopy images (observed under green and red optical filters) of pristine and functionalized SS surfaces after exposure to *E. coli* for 4 h. The viable cells appear green, while the dead cells appear red under the fluorescence microscope after staining with the combination dye. A clear difference in the number and physical state of *E. coli* cells on the pristine and modified SS surfaces is apparent. A high concentration of viable cells (stained green) can be seen on the pristine SS surface (Figure 4a), with very few dead cells (stained red) (Figure 4b), indicating that the bacteria are able to adhere and colonize the SS surface. For the SS-BC and SS-PDA surfaces (Figure 4c–f), a large number of viable cells and a few dead cells can be observed, indicating the poor ability of barnacle cement and polydopamine to resist bacterial adhesion. The results are consistent with the XPS results from antifouling assays.

For the SS-BC-PHEMA (Figure 5a,b) and SS-PDA-PHEMA surfaces (Figure 5c,d), both viable and dead cells were observed, but the total number of bacterial cells was significantly less than that observed on the pristine SS surface. The large reduction in bacterial adsorption demonstrates the resistance of PHEMA brushes to bacterial cell adhesion. For the SS-BC-CS (Figure 5e,f) and SS-PDA-CS surfaces (Figure 5g,h), the number of viable bacterial cells was substantially less than that observed for SS, and there was also an increase in the number of dead bacteria compared to what was observed on the pristine SS surface, indicating strong antibacterial ability of the chitosan-modified surface. Electrostatic interactions between the cationic sites of the chitosan-coupled chains and the anionic cell membrane can lead to leakage of proteins and other intracellular components, resulting in death of the bacterial cells.^{60–62} The sum of viable and dead cells on the SS-BC-CS (or SS-PDA-CS) surface was similar

to that of the corresponding SS-BC-PHEMA (or SS-PDA-PHEMA) surface, albeit with the presence of more dead cells on the chitosan-modified surface. The surfaces with the chitosan-coupled PHEMA brushes exhibit both antifouling and antibacterial properties. There was no marked difference between the SS-BC-PHEMA and SS-PDA-PHEMA surfaces in inhibiting bacterial adhesion or between the SS-BC-CS and SS-PDA-CS surfaces in terms of bactericidal activity.

To explore the killing efficiency of the functionalized surfaces further, a more quantitative *in vitro* antibacterial assay was carried out using the spread plate method. Figure 6 shows the relative viability of *E. coli* after exposure to the pristine and modified SS surfaces at 37 °C for 4 h. The poly(HEMA) and chitosan-functionalized SS substrate shows an obvious reduction in the number of adherent bacterial cells compared to the pristine SS surface. The viabilities of *E. coli* cells on the SS-BC-PHEMA and SS-PDA-PHEMA surfaces have decreased to 29.3% and 24.9%, respectively, of that of the cells on the pristine SS surface. The chitosan-functionalized substrates show the lowest relative viabilities of *E. coli*, with 18.7% and 16.8% for the SS-BC-CS and SS-PDA-CS surfaces, respectively. It has been reported that the number of viable *E. coli* decreases by 46–68% after 7 h of contact with the heparin/chitosan multilayer-modified poly(ethylene terephthalate),⁶³ while the number of viable *E. coli* decreases by 61.3% after 24 h of exposure of the quaternary ammonium salts-coated surface to the bacterial culture.⁶⁴ In comparison, the present bifunctional SS surfaces with PHEMA brushes and chitosan have shown better bacterial resistance and bactericidal properties, with lower relative viabilities of 16–19%. Thus, barnacle cement, extracted and utilized in its native state, is an effective initiator anchor with efficacy equal to that of dopamine.

3.6. Stability of Barnacle Cement and Dopamine on Stainless Steel Surfaces. Stability and durability of functional coatings are important criteria for evaluating long-term applications of materials in harsh environments. For the polymer brushes prepared by the “grafting-from” method described here, the stability of biomimetic anchors for immobilization of the polymerization initiator will have an important effect on the stability and durability of the functional polymer brushes on the substrate surfaces. To evaluate the stability of barnacle cement and dopamine as biomimetic anchors, the SS-BC and SS-PDA substrates were immersed in PBS solution at 37 °C for 30 days. Their surface composition was analyzed by XPS. As shown in Figure 7, the respective XPS wide scan, C 1s and N 1s core-level spectral line shapes of both the SS-BC surface (Figure 7a–f) and SS-PDA surface (Figure 7g–l) do not change appreciably after immersion in PBS for 30 days. XPS signals of the underlying metallic elements from SS (Fe 2p_{3/2} and Cr 2p_{3/2} core-level signals with BEs at around 711 and 577 eV, respectively) are discernible for the SS-BC and SS-PDA surfaces after the immersion test, indicating that the concentration (thickness) of barnacle cement and polydopamine on the respective SS surfaces has decreased slightly after 30 days in PBS. The $[N]/[C]$ ratios of the SS-BC surface are 0.24 and 0.22, respectively, before and after the immersion test. The C 1s core-level spectrum of the SS-BC surface after 30 days of immersion in PBS (Figure 7e) is comparable to that of the original SS-BC surface (Figure 7b) and can be similarly curve-fitted into five peak components with BEs at about 284.6, 285.6, 286.2, 287.8, and 288.9 eV, attributable to the C–H, C–N, C–O, N–C=O, and O–C=O species, respectively. The comparable $[N]/[C]$ ratios, as well as the C 1s (Figure 7b,e) and N 1s (Figure 7c,f) core-level spectral line shapes, of the SS-BC surface before and after 30 days of immersion in PBS indicate that the

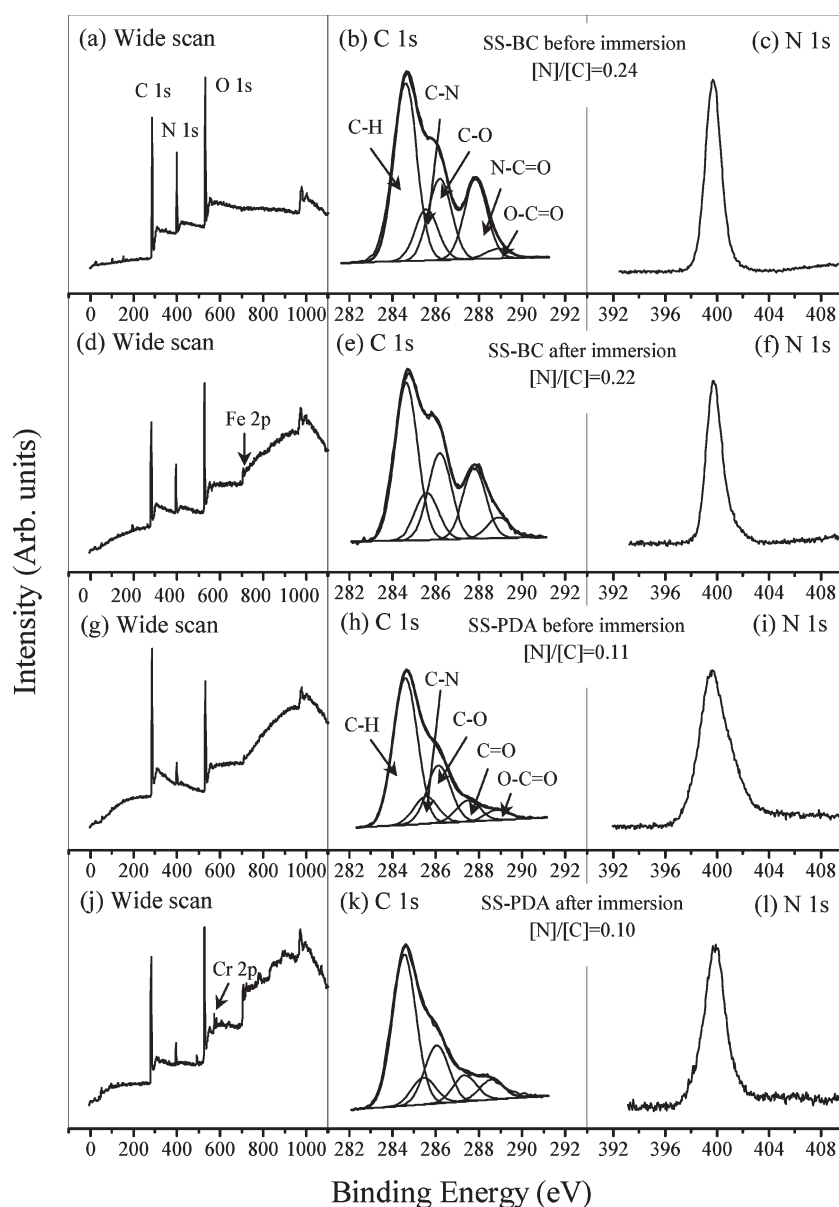


Figure 7. XPS wide scan, C 1s and N 1s core-level spectra of the (a, b, c) SS-BC surface before immersion in PBS solution, (d, e, f) SS-BC surface after immersion in PBS for 30 days, (g, h, i) SS-PDA surface before immersion in PBS solution, and (j, k, l) SS-PDA surface after immersion in PBS for 30 days.

composition of barnacle cement remains constant in PBS. The SS-PDA surface also exhibits good stability in PBS. The $[N]/[C]$ ratio decreases slightly from 0.11 to 0.10 after 30 days of immersion in PBS. The C 1s core-level spectra of the SS-PDA surface before and after immersion in PBS (Figure 7h,k) are also comparable and can be curve-fitted into five peak components with BEs at about 284.6, 286.2, 286.2, 287.4, and 288.9 eV, attributable to C–H, C–N, C–O, C=O, and O–C=O species, respectively, suggesting the retention of SS-PDA surface composition throughout the immersion process. Thus, both types of the biomimetic anchors are relatively stable and durable in aqueous media.

4. CONCLUSIONS

Barnacle cement (BC) can be beneficially utilized in its native state as a novel, biomimetic anchor for surface-initiated controlled radical polymerization to immobilize bifunctional (antiprotein and

antibacterial adhesion and bactericidal properties) polymer brushes on stainless steel (SS). To impart the antiadhesion and bactericidal properties, poly(2-hydroxyethyl methacrylate) (PHEMA) brushes were first grafted from the BC-modified SS surface via surface-initiated atom transfer radical polymerization of 2-hydroxyethyl methacrylate (HEMA), followed by coupling of chitosan. For comparison purpose, dopamine was also employed as a biomimetic anchor to prepare similarly functionalized SS surfaces. In comparison to the pristine SS, and the BC- and polydopamine-coated SS surfaces, the polymer brushes-functionalized SS surfaces using either type of biomimetic anchor effectively reduced protein adsorption and showed strong antibacterial efficacy against *E. coli*. The compositions of barnacle cement and dopamine remained constant after immersion in PBS for 30 days, with the loss of <10% of both adhesives from the respective surfaces, indicative of good stability and durability of both types of biomimetic anchors. In perhaps what is an ironic

twist, through proper surface molecular design, the outstanding adhesive properties of barnacle cement, which cause marine industries billions of dollars annually in fouling prevention, can be employed beneficially in the development of environmentally friendly antifouling and antibacterial surfaces. This result highlights the versatility of different classes of marine-derived cements for use in surface molecular design and extends the application ranges of biological adhesives. The present approach to achieve bifunctional (antifouling and antibacterial) stainless steel surfaces is potentially useful for the fabrication of biomedical devices and biomaterials.

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